Instrumentation and Control

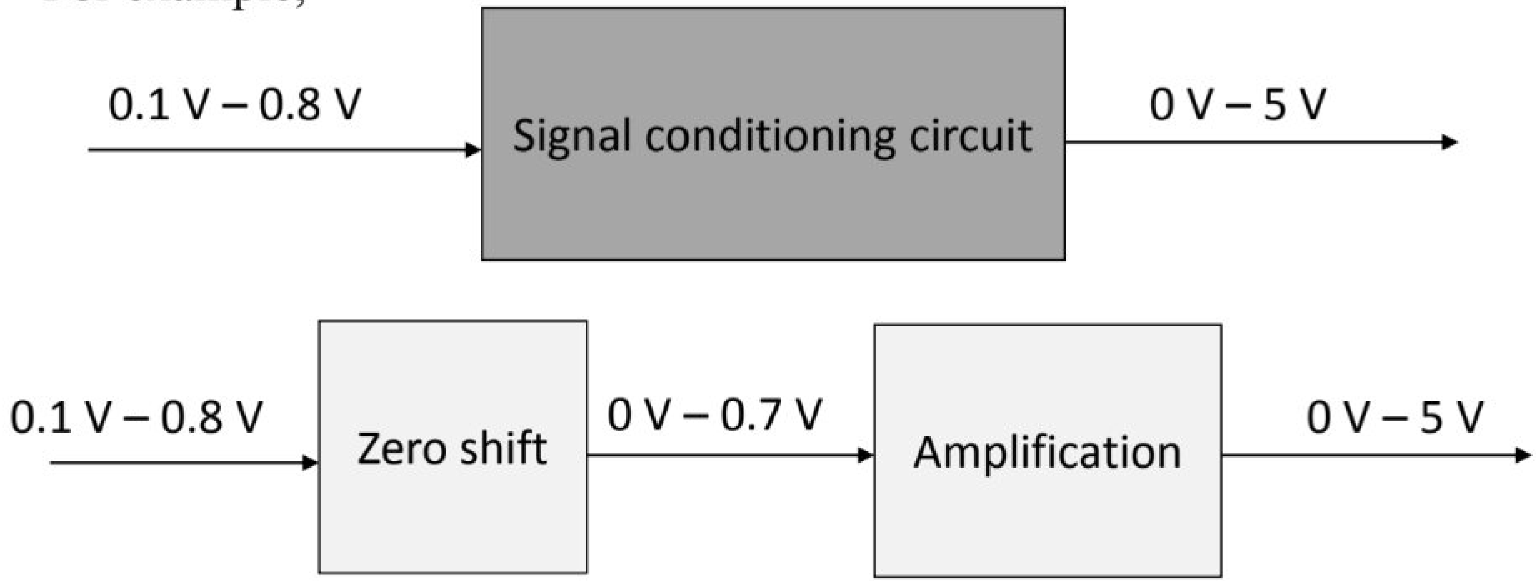
CSE 325

Md Atikul Islam
Lecturer(EEE)
Computer Science and Engineering(CSE)
Daffodil International University(DIU)

Change of signal level – getting the level of the signal right

It means adjusting the level (magnitude) and bias (zero value) of signal

- For example,



The signal might be just a few milivolts. To feed the signal into an A/D converter for inputting to a microprocessor, it needs to be made much larger; volts rather than milivolts. Operational amplifier circuits are widely used for amplification.

Introduction

- An amplifier is a device that amplifies a signal – almost always a voltage
- The low voltage output of a sensor, say of a thermocouple, may be amplified to a level required by a controller or a display.
- Amplification may be quite large – sometimes of the order of 10^6 or it may be quite small or even smaller than one, depending on the need of the sensor.
- Amplifiers can also be used for impedance matching purposes even when no amplification is needed.
- Power amplifiers, which usually connect to actuators, provides the power necessary to drive them.
- Amplifiers are sometimes incorporated in the sensor.

Amplifiers for Instrumentation

Desirable Characteristics of an Amplifier

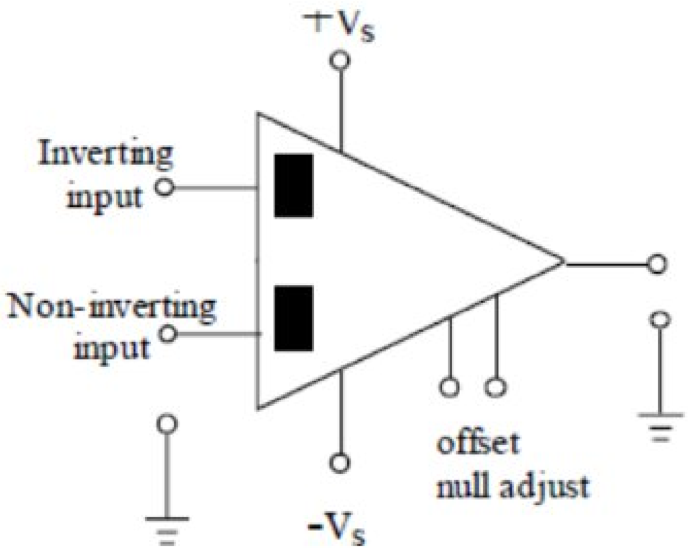
- The amplification factor should be constant (for linear scaling)
- The frequency response curve should be flat over the operating band
- The power extracted from the measurement system should be as small as possible Should not add noise to the sensor signal
- It should operate on the least possible auxiliary power
- It should have a long operating life and a high degree of reliability
- The amplifier should have a very high input impedance and very low output impedance. Why?

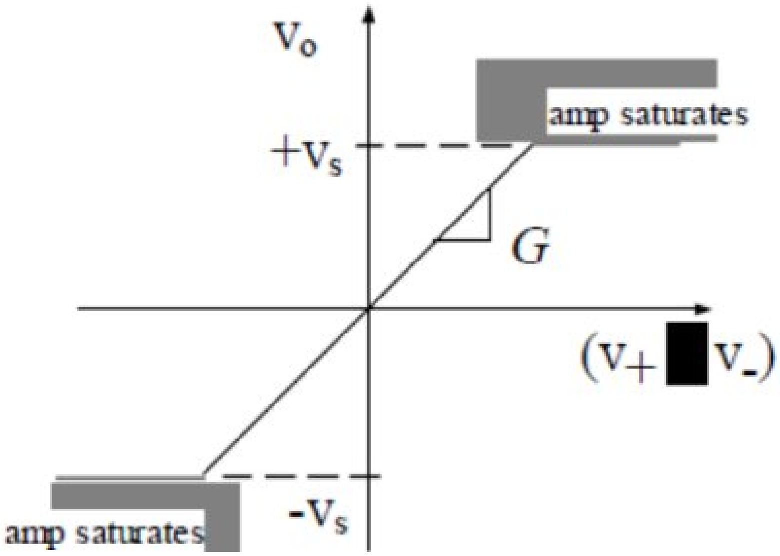
High Input Impedance (Z_{in})
Prevents the amplifier from **drawing current from the source** & Ensures **maximum voltage is transferred** from the source to the amplifier
Low Output Impedance (Z_{out})
Allows the amplifier to **deliver maximum current to the load** & Minimizes **voltage drop inside the amplifier**

Amplifiers for Instrumentation

Almost always, operational amplifiers (Op-amp) are used for amplification, as they have almost all the desirable characteristics mentioned earlier.

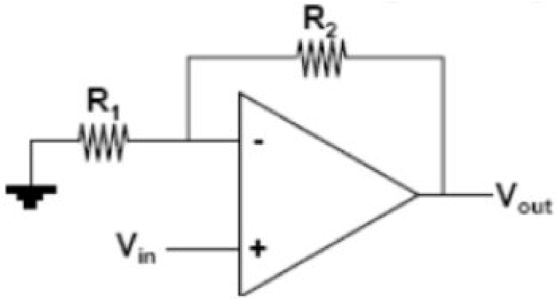
- Vary high gain, more than 106, ideally infinite
- Adjustable gain can be achieved by external circuit element
- High input impedance (ideally infinite) and low output impedance (ideally zero, practical values 20-100 Ω)





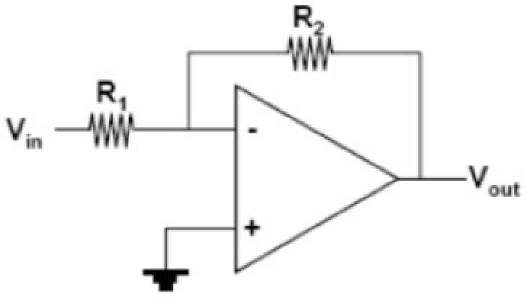
Circuit Configurations Using Op-Amps

Non-Inverting Operational Amplifier (Op-Amp)



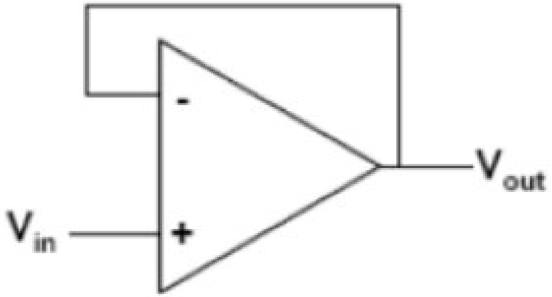
$$V_{out} = \left(1 + \frac{R_2}{R_1}\right)V_{in}$$

Inverting Operational Amplifier (Op-Amp)



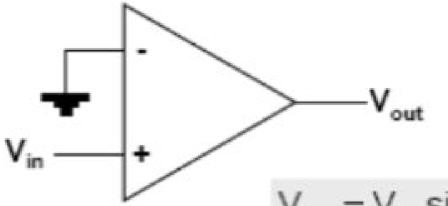
$$V_{out} = -\frac{R_2}{R_1}V_{in}$$

Voltage Follower (Buffer Amplifier)

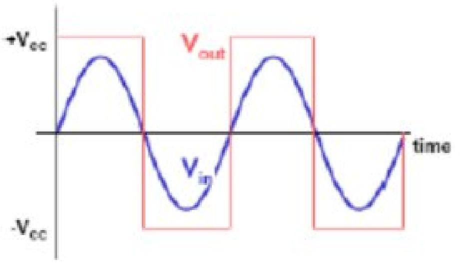


$$V_{out} = V_{in}$$

Voltage Comparator (Op-Amp Comparator)

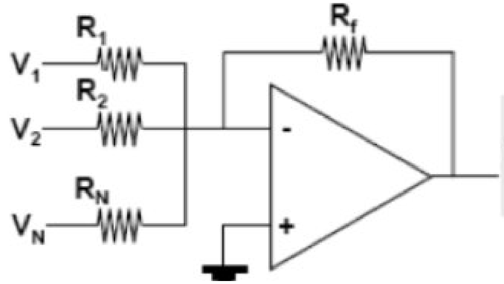


$$V_{out} = V_{CC}\,\mathrm{sign}(V_{in})$$



Circuit Configurations Using Op-Amps

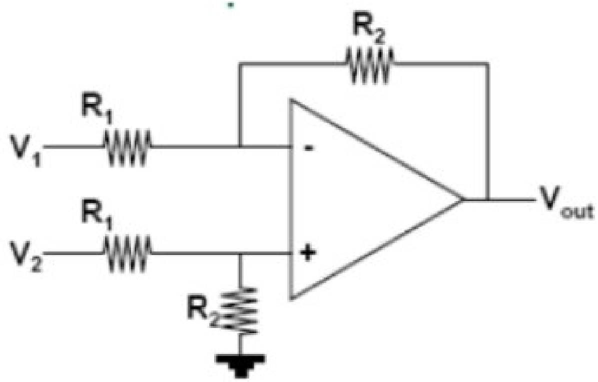
Summing Amplifier



$$V_{out} = -\left(V_1 \frac{R_f}{R_1} + V_2 \frac{R_f}{R_2} + \cdots + V_N \frac{R_f}{R_N} \right)$$

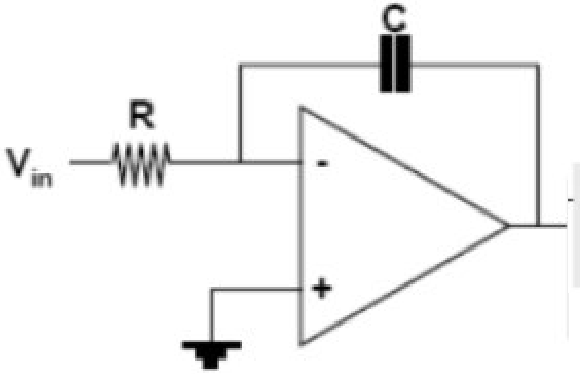
R_2

Differential Amplifier



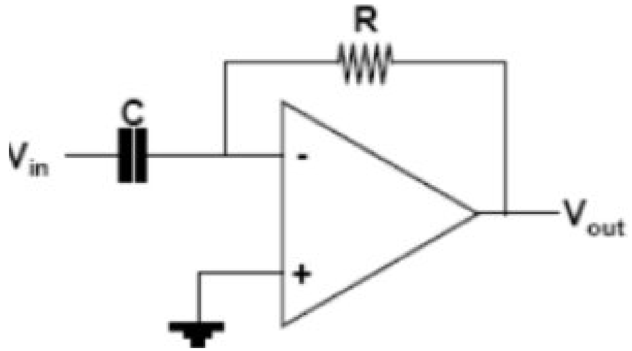
$$V_{out} = \frac{R_2}{R_1}(V_2 - V_1)$$

Integrating Amplifier (Integrator)



$$V_{out} = -\frac{1}{j\omega CR} V_{in} = -\frac{1}{RC}\int V_{in} dt$$

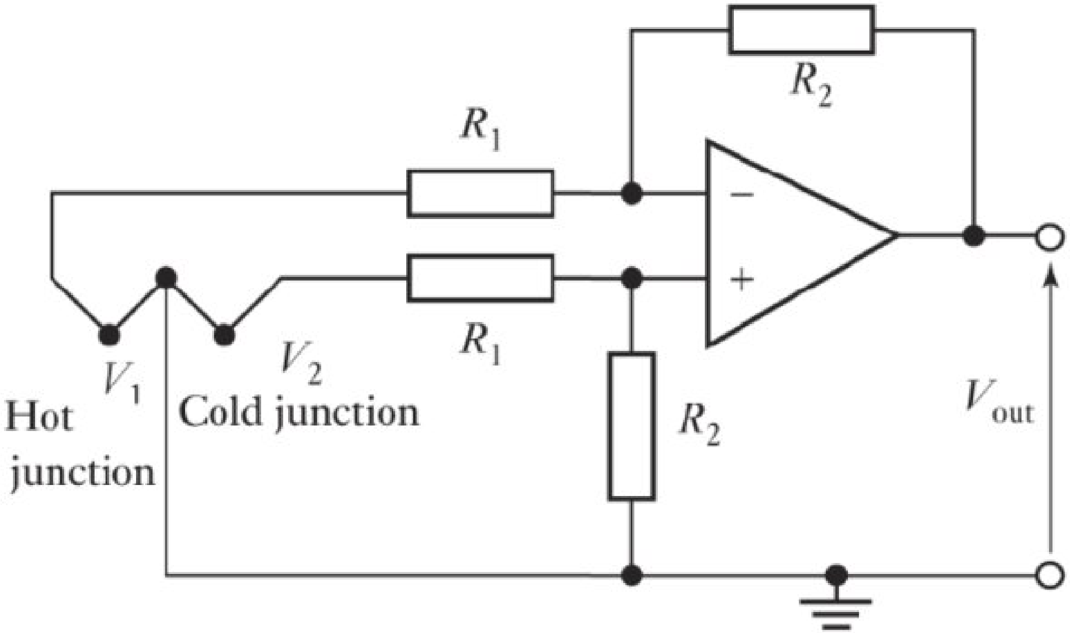
Differentiating Amplifier (Differentiator)



$$V_{out} = -\frac{R}{\frac{1}{j\omega C}} V_{in} = -RC \frac{dV_{in}}{dt}$$

Circuit Configurations Using Op-Amps

Example: The difference in the emfs of the two junctions of the thermocouple is being amplified. A temperature difference of 10 °C produces an emf difference of 530 μV. then the values of R1 and R2 can be chosen to give a circuit with an output of 10 mV.



$$V_{out} = \frac{R_2}{R_1}(V_2 - V_1)$$

$$10 \times 10^{-3} = \frac{R_2}{R_1} \times 530 \times 10^{-6}$$

$$\frac{R_2}{R_1} = 18.9$$

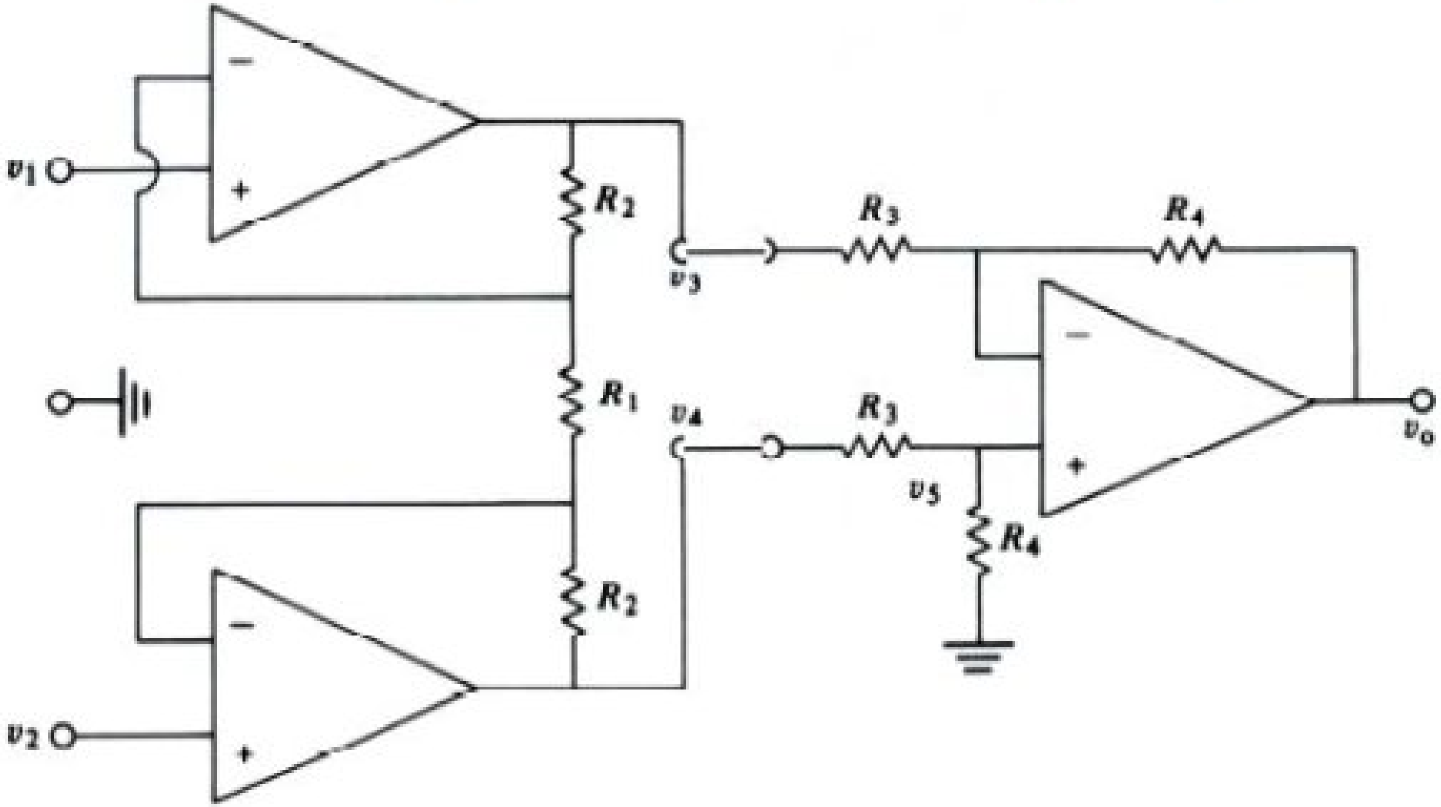
So if we select, R_1 as **10 k Ohm**

Then, $R_2 = 189$k Ohm

Instrumentation Amplifier

It is available in single IC and is designed to have:

- Very high input impedance (300 MΩ), very low output impedance
- High common-mode rejection ratio (more than 100 dB)
- High voltage gain



Total differential gain: $G_d = \dfrac{2R_2 + R_1}{R_1}\left(\dfrac{R_4}{R_3}\right)$

Math

Example & Practice Problem : 5.3, 5.5, 5.6, 5.7, 5.8.
Book: Fundamentals of Electrical circuits
Charles K.Alexander
Matthew N.O. Sadiku

Operational Amplifier- Properties

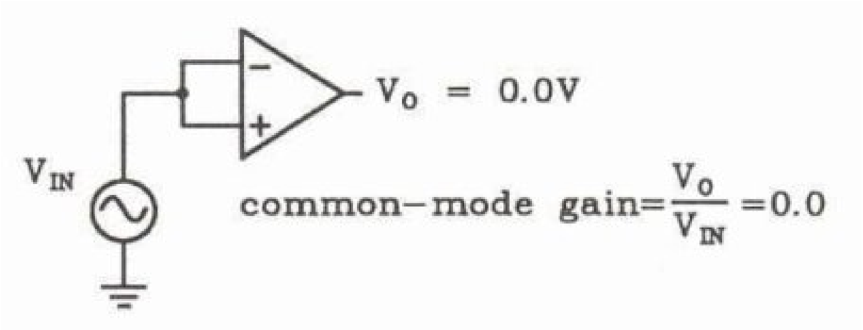
Differential voltage gain (A_d):

- the amplification of the op-amp of the difference between the two inputs.
- also called the open loop gain
- in a good amplifier it should be as high as possible.
- gains of 106 or higher are common.

- An ideal amplifier is said to have infinite gain.

Common-mode voltage gain (A_{cm}):

- By virtue of differential nature of the amplifier, this gain should be zero.
- Practical amplifiers may have a small common mode gain because of the mismatch between the two channels but this should be small.



Operational Amplifier- Properties

CMRR (Common Mode Rejection Ratio (CMRR)) is the ratio between A_d and A_{cm}:

$$CMRR = \frac{A_d}{A_{cm}}$$

It can also be expressed in dB

- In an ideal amplifier this is infinite.
- A good amplifier will have a CMRR that is very high

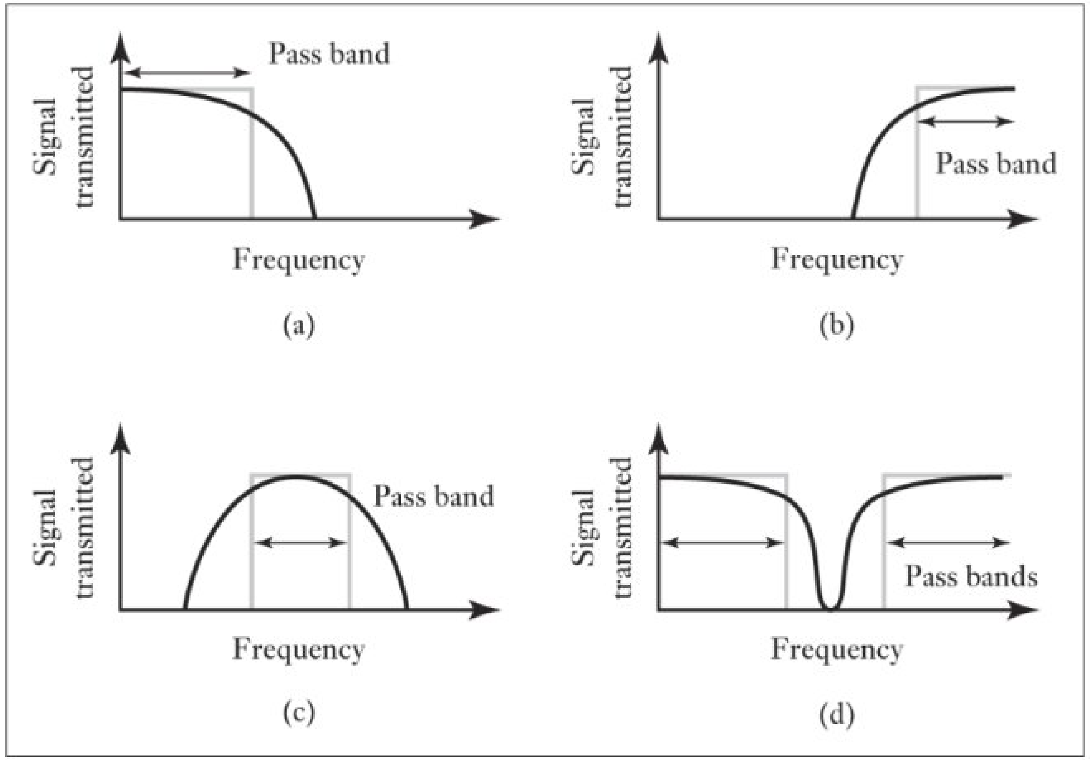
Eliminating or reducing noise – increasing the signal-to-noise ratio (SNR)

Noise may come from:

- ☔ Inherent noise (Internal to the sensor)
- ☔ Interference (external): main source of noise

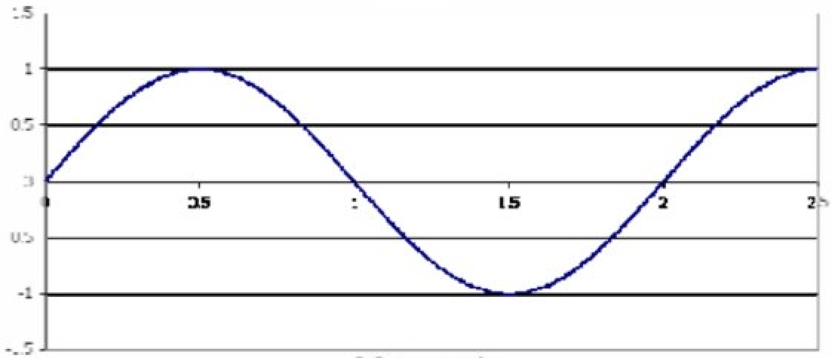
Filter or compensation circuits might be used to eliminate or reduce noise from a signal

Example filters:
(a) low-pass
(b) high-pass
(c) Band-pass
(d) Band-stop (notch filters)

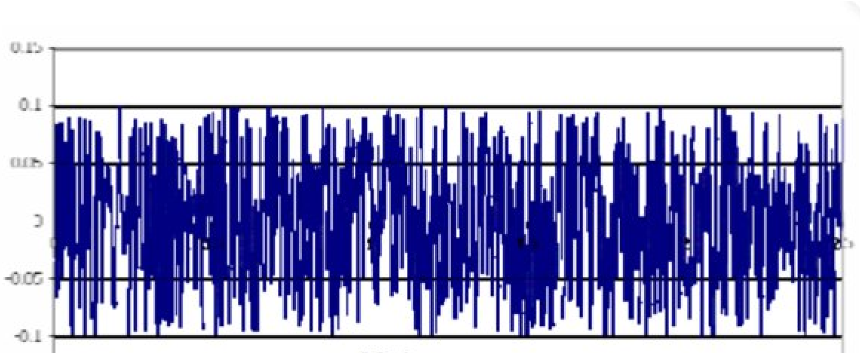


Noise : Introduction

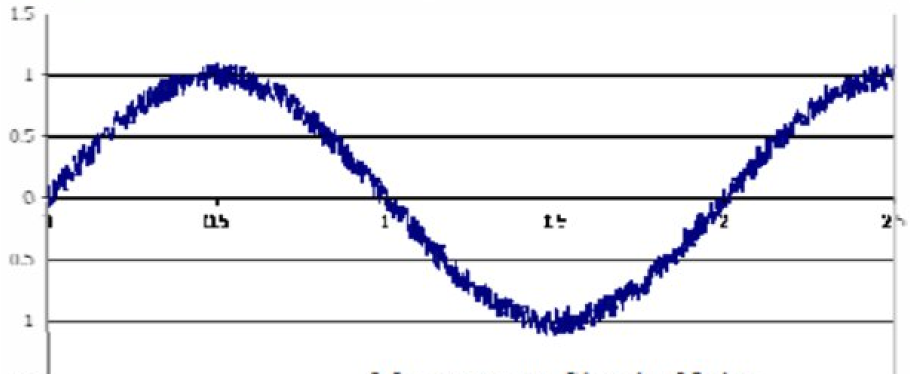
- Noise is a variation in the measurement of a variable that does not reflect real changes in the variable.
- Noise may come from different sources. Whatever the source, noise distorts the original signal



Measurand



Noise



Measurement: Signal + Noise

Effects of Noise

Effects of Noise

- Reduces precision, accuracy (sometimes) and resolution of the measurement.

- Introduces errors in control systems: To a controller, fluctuation in the process variable caused by noise are indistinguishable from fluctuations caused by real variable. Noise in a process variable will be reflected in the output of the controller.

Types and Sources of Noise

Many sources and many types of noise but it can be distinguished between two broad types

1. INTERNAL: Noise internal to a sensor/amplifier/circuit

2. EXTERNAL: Interference noise (external).

Internal Noise

Internal noise is generated due to many effects in the circuit elements. Some of them are avoidable, but some are intrinsic.

a) Thermal or Johnson noise

b) Shot noise

c) Pink noise

Thermal or Johnson noise:

One of the basic sources of the noises is the resistor R.

Causes: This is generated due to the chaotic thermal movement of the charges.

The noise power density is given by:

$$e_n^2 = 4kTR\Delta f \qquad \left[\frac{\mathrm{V}^2}{\mathrm{Hz}}\right]$$

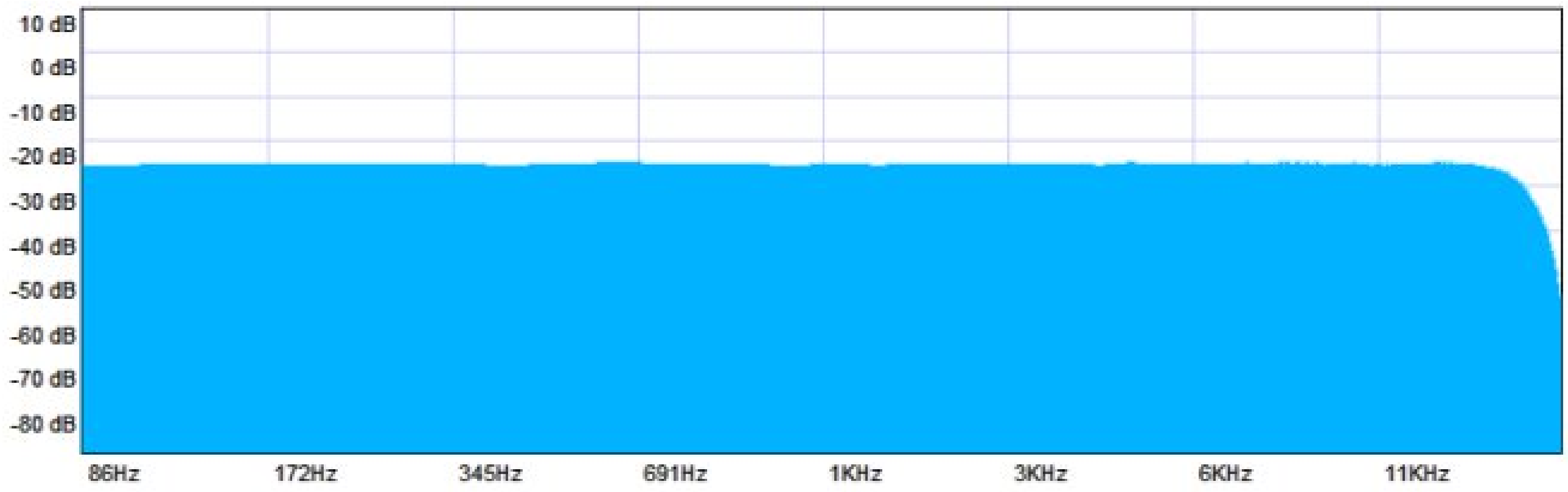
k is the Boltzman constant ($k = 1.38 \times 10^{-23}$ J/°K),
T is the temperature in °K,
R is the resistance in Ω
Δf is the bandwidth in Hz.

Solution:

- Can be reduced by the decrease of R, by limiting of bandwidth and of course by control of temperature.
- Fairly constant over wide range of frequencies. So, it is called a **WHITE NOISE**

WHITE NOISE

A white noise is a random signal with constant power spectral density over the entire frequency range.



Internal Noise

Shot noise:

This type of noise is produced in electronic/photonic devices with semiconductor junctions.

Causes: Produced in semiconductors when charges flows past the potential barrier by random collisions of electrons and atoms that causes fluctuations in the real signal current.

The shot noise I_{ss} described by the Schottky relation

$$I_{ss} = \sqrt{2qI\Delta f}$$

- q is the electronic charge, 1.6×10^{-19}C
- I is the DC current, Amp
- Δf is the bandwidth in Hz

Shot noise also is the white noise as it is fairly constant over a wide range of frequencies.

Internal Noise

Pink noise:

These are 1/f type noises (sometimes called low frequency noise, flicker noise or excess noise), as they are inversely proportional to the frequency.
So, unlike white noise, this type of noise has higher energy at low frequencies.

This is a problem for the sensors that operate at low frequencies.

There are many sources of these noises, most of them unknown.

Noise Evaluation

Noise is **random**, so we cannot describe it using a fixed amplitude like a normal signal.
Instead, we use **spectral density** $S(f)$,which tells us:

$$S(f) = \frac{V_n^2}{\Delta f}$$

•V_n^2 =noise power
•Δf= bandwidth
So, Spectral density = **noise power per unit bandwidth**

📊 2. Signal-to-Noise Ratio (SNR)

SNR tells us how strong the signal is compared to noise:

$$SNR = \frac{signal\ power}{noise\ power}$$

Higher SNR = better signal quality
Lower SNR = noisy signal

📏 3. SNR in Decibels (dB)

$$(SNR)_{dB} = 20\log_{10}\left(\frac{V_s}{V_n}\right)$$

•V_s =signal voltage
•V_n =noise voltage